Online appendix for "Endogenous Plucking Through Networks: The Plucking Paradox"

Contents

A Derivations for the static equilibrium

This appendix derives the static equilibrium conditions reported in Section 2. We assume perfect competition in the simple sector and monopolistic competition for varieties sold to the retailer. All firms are symmetric within each sector.

Marginal costs. The network firm minimizes the cost of the Cobb–Douglas aggregator in the main text. Let λ^N denote unit cost for a network firm and λ^S the unit cost for a simple firm. With competitive input prices and symmetry,

$$\ln \lambda^N = \alpha \ln \lambda^{N,\mathrm{agg}} + (1-\alpha) \ln \lambda^{S,\mathrm{agg}} - \ln A, \tag{1}$$

$$\ln \lambda^S = \ln w - \ln A. \tag{2}$$

Prices. Equating $\lambda^S, \lambda^N, \alpha$ in (2) with $\lambda^{S,\mathrm{agg}}, \lambda^{N,\mathrm{agg}}, \alpha^{\mathrm{agg}}$ since the model features representative firms, we can solve for the sectoral prices as functions of A and w

$$\ln \lambda^{N,\mathrm{agg}} = \ln w - \frac{2-\alpha^{\mathrm{agg}}}{1-\alpha^{\mathrm{agg}}} \ln A, \tag{3}$$

$$\ln \lambda^{S,\mathrm{agg}} = \ln w - \ln A. \tag{4}$$

Retailer's cost minimization. The retailer faces constant-elasticity demand within each block, so the sectoral price indices satisfy

$$P = (p^{N,\mathrm{agg}})^{\zeta} (p^{S,\mathrm{agg}})^{1-\zeta} = \left(\frac{\sigma}{\sigma-1}\right)^{\zeta} (\lambda^{N,\mathrm{agg}})^{\zeta} (\lambda^{S,\mathrm{agg}})^{1-\zeta}. \tag{5}$$

Normalizing $P \equiv 1$ and substituting (1)–(2) yields

$$\ln w = \left(\frac{\zeta}{1-\alpha^{\mathrm{agg}}} + 1\right) \ln A - \zeta \ln \frac{\sigma}{\sigma-1}, \tag{6}$$

which corresponds to the wage equation in the text.

Accounting and profits. Let TC denote total cost of network intermediates used by network firms. A fraction αTC is network goods and $(1-\alpha)TC$ is simple goods. Total cost of simple goods is wL, so the retailer spends $wL-(1-\alpha)TC$ on simple goods and, by CES expenditure shares, $\frac{\zeta}{1-\zeta}(wL-(1-\alpha)TC)$ on network goods. Converting retailer spending on network goods into costs using the sectoral markup implies

$$\alpha TC + \frac{\zeta}{1-\zeta}\frac{\sigma-1}{\sigma}\big(wL-(1-\alpha)TC\big) = TC, \tag{7}$$

so

$$TC = \frac{\zeta(\sigma-1)}{(1-\alpha)(\sigma-\zeta)}wL. \tag{8}$$

Retailer spending on network goods is then $\frac{\sigma\zeta}{\sigma-\zeta}wL$, and network profits equal revenue minus cost,

$$\pi^N = \frac{\zeta}{\sigma-\zeta}wL, \tag{9}$$

while perfect competition in the simple sector implies

$$\pi^S = 0. \tag{10}$$

Finally, total consumption equals total revenue,

$$C = \frac{\sigma}{\sigma-\zeta}wL = wL + \pi^N, \tag{11}$$

which matches the aggregate consumption expression in the main text.

Firm-specific profits. Let $X = [\alpha^{\text{agg}}, A]$ and consider a firm that chooses its own network intensity α while taking aggregate prices as given. Using (1) with aggregate prices (3)–(4) and (6), the firm's unit cost satisfies

$$\ln \lambda^N(\alpha; X) = \left(\frac{\zeta - \alpha}{1 - \alpha^{\text{agg}}} - 1\right) \ln A - \zeta \ln \frac{\sigma}{\sigma - 1}, \tag{12}$$

The aggregate $\lambda^{N,\text{agg}}(X)$ satisfies

$$\ln \lambda^{N,\text{agg}}(X) = \left(\frac{\zeta - 1}{1 - \alpha^{\text{agg}}}\right) \ln A - \zeta \ln \frac{\sigma}{\sigma - 1}. \tag{13}$$

Under CES demand, revenue is proportional to $(p(\alpha; X)/p^{N,\text{agg}}(X))^{1-\sigma}$, implying

$$\begin{aligned}
\pi^N(\alpha; X) &= \pi^{N,\text{agg}}(X) \left(\frac{p(\alpha; X)}{p^{N,\text{agg}}(X)}\right)^{1-\sigma} && (14)\\
&= \pi^{N,\text{agg}}(X) \left(\frac{\lambda^N(\alpha; X)}{\lambda^{N,\text{agg}}(X)}\right)^{1-\sigma} && (15)\\
&= \frac{\zeta}{\sigma - \zeta} \left(\frac{\sigma - 1}{\sigma}\right)^{\zeta} A^{\frac{\zeta + (1-(1-\delta)\alpha)(1-\sigma)}{1-(1-\delta)\alpha^{\text{agg}}} + \sigma} L, && (16)
\end{aligned}$$

which is consistent with the main text.

B Derivations for the optimal fiscal policy

This appendix derives the tax rule that decentralizes the planner allocation by equating the competitive-equilibrium (CE) and social-planner (SPP) Euler equations. Throughout this section, we work with the dynamic model: the effective network intensity entering production is $(1 - \delta)\alpha$, where α is the previous period's choice (replacing α^{agg} in the static expressions of Appendix A), and labor supply $n(X)$ is endogenous under GHH preferences (replacing the inelastic supply L).

Competitive Equilibrium Euler Equation. Let $\tilde{c}(X)$ denote household consumption in state X. The firm Euler in CE with a proportional tax/subsidy τ on network profits is

$$\mathbb{E}\left[\beta\frac{(\tilde{c}'(X'))^{-\rho}}{(\tilde{c}(X))^{-\rho}}\left(\pi_1(\alpha';X')(1+\tau')-\Phi_1(\alpha',\alpha'')\right)\Big|X\right]=\Phi_2(\alpha,\alpha')-\lambda_{CE}+\mu_{CE}, \quad (17)$$

where, using the static equilibrium,

$$w(X)=\left(\frac{\sigma-1}{\sigma}\right)^{\zeta}A^{\frac{\zeta}{1-(1-\delta)\alpha}+1}, \quad (18)$$

$$n(X)=\left(\frac{w(X)}{\eta}\right)^{\chi}=\eta^{-\chi}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta\chi}A^{\left(\frac{\zeta}{1-(1-\delta)\alpha}+1\right)\chi}, \quad (19)$$

$$\pi(\alpha;X)=\frac{\zeta}{\sigma-\zeta}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta}A^{\frac{\zeta}{1-(1-\delta)\alpha}+1}n(X). \quad (20)$$

Thus

$$\pi_1(\alpha;X)=\pi(\alpha;X)\,\log(A)\,\frac{(\sigma-1)(1-\delta)}{1-(1-\delta)\alpha}. \quad (21)$$

$$=\frac{\zeta}{\sigma-\zeta}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta}A^{\frac{\zeta}{1-(1-\delta)\alpha}+1}\eta^{-\chi}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta\chi}A^{\left(\frac{\zeta}{1-(1-\delta)\alpha}+1\right)\chi}\log(A)\,\frac{(\sigma-1)(1-\delta)}{1-(1-\delta)\alpha}, \quad (22)$$

$$=\frac{\zeta}{\sigma-\zeta}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta(1+\chi)}\eta^{-\chi}A^{\left(\frac{\zeta}{1-(1-\delta)\alpha}+1\right)(1+\chi)}\log(A)\,\frac{(\sigma-1)(1-\delta)}{1-(1-\delta)\alpha}, \quad (23)$$

Planner's Euler Equation. The SPP Euler equation is

$$\beta\mathbb{E}\Bigg[(\tilde{c}')^{-\rho}\Bigg(\left(\frac{\sigma}{\sigma-\zeta}\right)^{1+\chi}\left(\frac{\sigma-1}{\sigma}\right)^{\zeta(1+\chi)}\eta^{-\chi}(A')^{\left(\frac{\zeta}{1-(1-\delta)\alpha'}+1\right)(1+\chi)}\log(A')\,\frac{\zeta(1-\delta)}{(1-(1-\delta)\alpha')^2}$$
$$-\Phi_1(\alpha',\alpha'')\Bigg)\Bigg]$$
$$=(\tilde{c})^{-\rho}\Phi_2(\alpha,\alpha')-\lambda_{SP}+\mu_{SP}. \quad (24)$$

Tax rule. Equating (17) and (24) (and matching multipliers via $(\tilde{c})^{\rho}\lambda_{SP} = \lambda_{CE}$ and $(\tilde{c})^{\rho}\mu_{CE} = \mu_{SP}$) yields

$$1 + \tau' = \frac{\sigma - \zeta}{\sigma - 1}\left(\frac{\sigma}{\sigma - \zeta}\right)^{1+\chi}\frac{1}{1 - (1-\delta)\alpha'} \tag{25}$$

Budget balance. The subsidy is rebated lump-sum to households. Using $c = wn + \pi(1+\tau) - \Phi(\alpha, \alpha') - \tau\pi$, the resource constraint reduces to

$$c = wn + \pi - \Phi(\alpha, \alpha'), \tag{26}$$

so the tax leaves aggregate resources unchanged.

C A plucking-cycle exercise in simulated output

This appendix reproduces, in the simulated model, the plucking-cycle exercise in Dupraz, Nakamura, and Steinsson (2025). The exercise asks whether the model-generated output series displays the one-sided predictability pattern emphasized in the plucking literature: contraction amplitudes predict subsequent recoveries, while recovery amplitudes have little predictive power for subsequent contractions.

Following Dupraz, Nakamura, and Steinsson (2025), we apply their threshold turning-point algorithm to simulated model output. Their empirical implementation identifies peaks and troughs in unemployment in levels. Since the algorithm is designed for unemployment, where higher values indicate greater slack, we transform simulated output into a comparable shortfall measure. Consistent with Friedman's ceiling interpretation, we measure output relative to a smooth moving ceiling and define

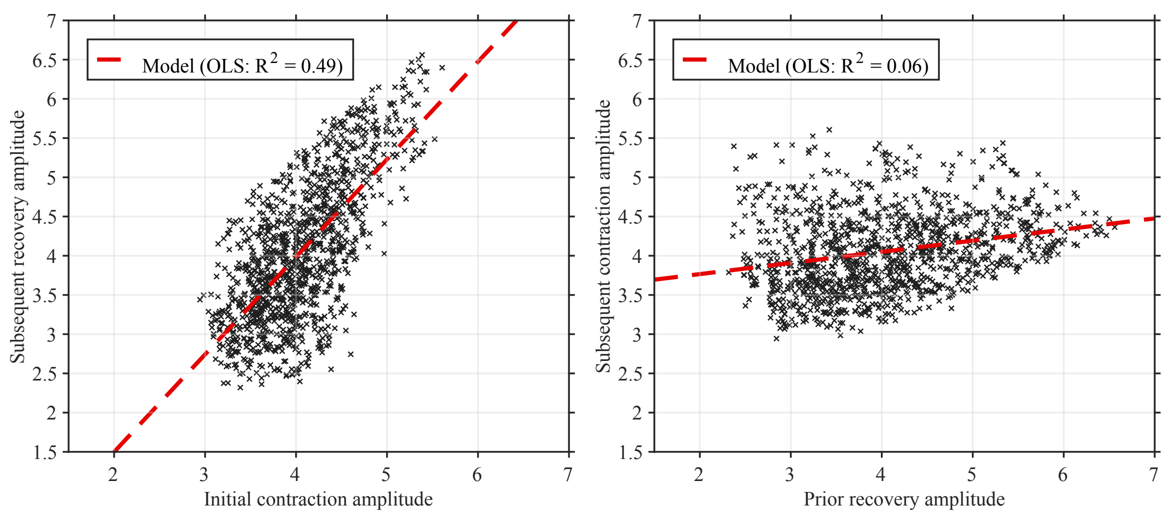
$$u_t = -\widehat{y}_t, \qquad \widehat{y}_t = \log Y_t - \mathrm{HP}(\log Y_t),$$

where the HP smoothing parameter is $\lambda = 6.25$ for annual data. We then apply the same threshold turning-point logic to u_t, with the cycle-size threshold set to 1.5

standard deviations of u_t. For each identified cycle, we compute the initial contraction amplitude and the subsequent recovery amplitude from the output-gap series $\widehat{y}_t$.

Figure C.1 reports the two plucking regressions. The left panel regresses the subsequent recovery amplitude on the initial contraction amplitude. The right panel regresses the subsequent contraction amplitude on the prior recovery amplitude. In the simulated network economy, contractions strongly predict subsequent recoveries, with an R^2 of 0.49. In contrast, prior recoveries explain little of the next contraction, with an R^2 of 0.06. This one-sided predictability is consistent with the model's plucking interpretation: downturns represent drops below a moving ceiling, and recoveries tend to close the previous gap, while the next downturn is governed by the new state of the economy rather than mechanically by the size of the previous recovery.

Figure C.1: Plucking-cycle exercise for simulated output slack



Notes: The figure applies the turning-point algorithm in Dupraz, Nakamura, and Steinsson (2025) to simulated model output slack, $u_t = -\widehat{y}_t$, where $\widehat{y}_t$ is the HP-filtered log output gap. The cycle-size threshold is 1.5 standard deviations of u_t. Amplitudes are measured as percentage-point changes in the output gap. The left panel regresses subsequent recovery amplitudes on initial contraction amplitudes. The right panel regresses subsequent contraction amplitudes on prior recovery amplitudes.

D Additional empirical results

In this section, we present additional empirical results that further support the model's mechanism and its implications for sectoral employment dynamics during recessions. Table D.1 reports the results from the regression of the depth of employment contraction on material share (with energy cost included), controlling for various industry characteristics. The coefficient on material share remains positive and statistically significant across all specifications, reinforcing the finding that industries with higher material intensity experience deeper contractions during recessions. Table D.2 presents the results from the trough-level regressions, showing that industries with higher material share (with energy cost included) not only experience deeper contractions but also reach lower levels of employment and total payroll at the trough. This further corroborates the model's mechanism, as a more network-intensive production structure exacerbates the depth of downturns, leading to more pronounced declines in key economic indicators during recessions.

Table D.1: Material share (including energy) and peak-to-trough drop [Data]

Drop^Y, Y=	(1) emp	(2) pay	(3) emp	(4) pay	(5) emp	(6) pay
Material Cost Share	0.132 (0.039)	0.140 (0.042)			0.125 (0.041)	0.123 (0.043)
Booming Duration			0.014 (0.002)	0.011 (0.002)	0.014 (0.002)	0.011 (0.002)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.349	0.352	0.363	0.365	0.365	0.367
N	5882	5882	5518	5518	5518	5518

Notes: Dependent variable is the peak-to-trough drop in employment, payroll, value added. Material share is measured at the peak year (pre-determined). Industry and peak-year fixed effects are included. Capital stock and the outcome variable at the peak level is included as control; standard errors are clustered by industry. Standard errors in parentheses.

Table D.2: Trough level regression (including energy cost in material cost) [Data]

Drop^{Y}, Y=	(1) emp	(2) pay	(3) emp	(4) pay	(5) emp	(6) pay
Material Cost Share	-0.131	-0.141			-0.122	-0.123
	(0.039)	(0.043)			(0.041)	(0.044)
Booming Duration			-0.014	-0.011	-0.014	-0.011
			(0.002)	(0.002)	(0.002)	(0.002)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.989	0.992	0.989	0.991	0.989	0.991
N	5882	5882	5518	5518	5518	5518

Notes: Dependent variable is the drop in employment, payroll. Material share is measured at the peak year (pre-determined). Industry and peak-year fixed effects are included. Capital stock at the peak level is included as control; standard errors are clustered by industry. Standard errors in parentheses.

E Canonical RBC model for benchmark

This appendix describes the canonical RBC model used to construct the benchmark conditional saddle paths in Figure 5(a). A representative household solves

$$v(a; K, A) = \max_{c,a'} \quad \log(c) + \beta \mathbb{E}\left[v(a'; K', A') \mid A\right] \tag{27}$$

$$\text{s.t.} \quad c + a' = a(1 + r(K, A)) + w(K, A) \tag{28}$$

$$a' \geq 0, \tag{29}$$

where v is the value function, a is wealth at the beginning of the period, K is aggregate capital, and A is aggregate TFP. The aggregate state evolves according to

$$K' = \Gamma_{\text{endo}}(K, A), \qquad A' \sim \Gamma_{\text{exo}}(\cdot \mid A), \tag{30}$$

where Γ_{endo} is the equilibrium law of motion for capital. TFP follows a two-state Markov chain between $A \in \{B, G\}$ with $G > B$ and transition matrix

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix}. \tag{31}$$

Production is Cobb-Douglas with constant returns to scale. Competitive factor prices are

$$r(K, A) = \alpha A K^{\alpha-1} - \delta, \tag{32}$$

$$w(K, A) = (1 - \alpha) A K^{\alpha}, \tag{33}$$

where δ is the depreciation rate. In equilibrium, the representative household's asset holdings equal the aggregate capital stock: $a = K$.

References

Dupraz, Stéphane, Emi Nakamura, and Jón Steinsson. 2025. “A plucking model of business cycles.” *Journal of Monetary Economics* 152:103766.